

Computational Analysis of Newtonian vs Relativistic Gravitational Potentials: Error Quantification and Machine Learning Approximations

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1 Abstract

Understanding the applicability limits of classical gravitational models and identifying the conditions under which relativistic corrections become significant remain important challenges in computational physics. In this study, a static relativistic gravitational potential is investigated through an integrated framework combining analytical analysis, numerical evaluation, and machine-learning-based approaches.

The relativistic formulation is analyzed and compared with the corresponding Newtonian approximation through mathematical derivation and computational error analysis. The deviation between the two models is evaluated through relative error analysis, enabling a systematic investigation of the transition between weak-field and strong-field gravitational regimes.

To explore the capability of data-driven methods in approximating nonlinear physical systems, a computational dataset is generated from the analytical model and used to train multiple machine learning algorithms. Linear regression, polynomial regression, neural networks, and random forest regression are evaluated to compare their ability to reproduce the behavior of the relativistic potential.

The results demonstrate that machine learning models can effectively approximate the underlying physical relationship, with nonlinear approaches providing significantly improved performance over simpler regression methods. The predictions are interpreted as computational approximations that complement the analytical formulation rather than replacing the physical model.

This study presents a framework that combines theoretical physics, numerical analysis, and machine learning for studying nonlinear scientific systems. The

approach highlights the potential of machine learning as a tool for computational modeling while maintaining consistency with established physical principles.

Keywords: Gravitational potential; Relativistic gravity; Newtonian approximation; Error analysis; Schwarzschild spacetime; Machine learning; Random forest regression; Computational physics

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2 Introduction

2.1 Motivation

Mathematical modeling provides a fundamental framework for describing physical systems by establishing quantitative relationships between observable quantities and theoretical predictions. Through appropriate mathematical formulations, complex physical phenomena can be analyzed, simulated, and compared under different conditions.

Gravitational systems represent an important example where the validity and limitations of different theoretical descriptions must be carefully examined. Newtonian gravity has successfully explained a wide range of phenomena and remains highly accurate in weak gravitational fields. However, its applicability becomes limited in regimes where spacetime curvature and relativistic effects become significant [1]. In such situations, general relativity provides the appropriate theoretical framework for obtaining accurate descriptions of gravitational behaviour [2].

Understanding the transition between classical and relativistic descriptions is an important problem in computational physics because it requires quantifying the deviation between different theoretical models. Such analysis provides a systematic method for determining the range of validity of classical approximations and identifying the regimes where relativistic corrections become necessary.

In this study, this problem is investigated through a combined analytical, numerical, and machine learning based approach. The aim is not to modify the underlying physical formulation, but to examine how computational techniques, including machine learning methods, can be used as approximation tools for a theoretically defined gravitational system.

2.2 Background of Relativistic Potential Models

The mathematical description of gravitational interactions has evolved from classical formulations based on Newtonian mechanics to relativistic models based on the geometry of spacetime [1, 2]. In Newtonian mechanics, gravitational effects are represented through forces and associated potential functions, which provide an effective description of the system under appropriate assumptions.

The Newtonian potential provides an accurate approximation when the gravitational field is sufficiently weak and spacetime curvature effects can be neglected. However, as the strength of the gravitational field increases, deviations from the classical approximation become significant, requiring a relativistic description.

General relativistic models introduce corrections arising from the interaction between matter, energy, and spacetime geometry [2]. Among the important solutions of Einstein's field equations, the Schwarzschild solution provides a fundamental description of the gravitational field produced by a static spherically symmetric mass distribution [3]. The difference between Newtonian and relativistic descriptions becomes increasingly important near regions of strong

gravitational fields.

Quantifying this difference provides insight into the applicability limits of classical gravitational models. By evaluating the deviation between the two descriptions, it becomes possible to determine the regions where Newtonian approximations remain valid and where relativistic corrections significantly influence the predicted behaviour.

In this work, a static relativistic potential model is investigated using analytical derivation and computational analysis [3]. The corresponding Newtonian approximation is examined, and the deviation between the two models is quantified through error analysis. Furthermore, machine learning based approximations are developed to investigate whether data-driven methods can effectively reproduce the behaviour of the relativistic potential.

2.3 Computational Physics and Machine Learning

Computational methods have become an essential component of modern scientific research because they allow physical systems to be explored beyond the limitations of purely analytical approaches. Numerical techniques enable systematic investigation of theoretical models, parameter variations, and complex relationships between physical quantities.

However, repeated evaluation of mathematical models can become computationally demanding, particularly when large parameter spaces or multiple simulations are involved. This has motivated the development of alternative computational approaches capable of learning and approximating complex relationships from generated or observed data.

Machine learning provides a framework for constructing predictive models by identifying patterns within datasets. In scientific applications, machine learning methods are increasingly used as surrogate models, where computationally efficient approximations are developed while maintaining consistency with established physical relationships [4, 5].

The application of machine learning in physics requires careful validation against theoretical expectations because high predictive accuracy alone does not guarantee physical reliability. Therefore, the combination of analytical modelling, numerical analysis, and machine learning evaluation provides a more complete approach for studying scientific systems.

In this research, machine learning methods are applied to the static relativistic potential model to investigate their ability to approximate a known nonlinear physical relationship. The machine learning component is considered as a complementary computational approach rather than a replacement for the analytical formulation.

2.4 Objectives and Contributions

The primary objective of this study is to develop a computational framework that integrates theoretical gravitational modelling, numerical error analysis, and

machine learning based approximation for analysing a static relativistic potential system.

The study focuses on understanding the relationship between the relativistic and Newtonian descriptions of gravitational behaviour and evaluating whether machine learning models can accurately reproduce the resulting nonlinear relationship.

The major objectives of this research are:

1. Developing the mathematical formulation of the static relativistic potential and its corresponding Newtonian approximation.
2. Quantifying the deviation between the relativistic and classical models using analytical and numerical error analysis.
3. Investigating the transition between strong-field and weak-field regimes through the behaviour of the relative error.
4. Constructing a computational dataset generated from the theoretical model for machine learning experiments.
5. Evaluating multiple machine learning architectures, including regression models, neural networks, and ensemble-based methods.
6. Comparing model performance using standard regression evaluation metrics and analysing their ability to approximate the physical relationship.

The main contribution of this work is the integration of three complementary components: analytical gravitational modelling, numerical quantification of model deviation, and machine learning based approximation. Rather than considering artificial intelligence as an alternative to theoretical physics, this study investigates its role as a computational tool for representing known physical relationships.

2.5 Organization of the Paper

The remainder of this paper is organized as follows. Section 2 presents the theoretical framework of the static relativistic potential and its Newtonian approximation. Section 3 describes the mathematical formulation and derivation of the error function. Section 4 presents the numerical analysis of the relativistic and classical models. Section 5 discusses computational implementation and dataset generation. Section 7 introduces the machine learning framework and describes the evaluated models. Section 8 presents the model performance results and analysis. Finally, Section 9 discusses the physical interpretation, limitations, and future extensions of the study.

3 Theoretical Framework

3.1 Physical Motivation

Mathematical models describing physical systems need to be accurate and reflect correctly the physical phenomena occurring within such systems depending on various conditions. While classical gravitational systems give reliable predictions in many cases, their limitations arise when dealing with strong enough gravitational fields and relativistic effects.

Comparison of different descriptions gives insight into the boundaries of applicability of one or another physical model. The comparison of the deviation between such models enables us to quantify the transition from one regime to another.

In the case of the weak-field approximation, the Newtonian approximation gives a good description, while relativistic corrections can be neglected. As the characteristic distance from the gravitational field decreases, relativistic corrections start to play a bigger role and the classical description becomes less precise.

The aim of this study is to examine the process of the transition from the classical to the relativistic description by constructing a computational framework which allows us to compare the relativistic and the Newtonian potentials.

By integrating analytical methods, numerical simulations, and machine learning algorithms, this study investigates both the underlying physical behaviour of the system and the feasibility of developing accurate computational approximations.

The theoretical framework proposed in this section serves as a basis for the subsequent mathematical description, analysis of errors and machine learning experiments.

3.2 Dimensionless Representation

In order to facilitate the mathematical manipulation of the model and generalize the outcome to various physical scales, the gravitational field is expressed in terms of a dimensionless variable.

The normalization of the radius vector is done using the characteristic length of the relativistic gravitational system:

$$x = \frac{r}{r_s}$$

where:

- r represents the radial distance from the gravitational source.
- r_s represents the characteristic relativistic scale of the system.
- x represents the normalized dimensionless radial parameter.

Dimensional variable gives several benefits. First, it eliminates dependence on the specific physical units and allows the system to be studied in terms of the relative distance instead of the absolute one.

The variable (x) can be also used for the examination of various gravitational regimes. The values of the variable (x) that are close to the characteristic length correspond to the relativistic gravitational systems. Larger values of the variable (x) correspond to gravitational systems with low intensity where classical approximation is a better choice.

In terms of this normalized distance both the relativistic and Newtonian potential models can be expressed, and thus the comparison between them will not depend on any specific physical system but will rely only on mathematics.

This dimensionless formulation forms the basis for the analytical derivations and computational simulations performed in the following sections.

3.3 Newtonian Approximation

The Newtonian gravitational potential provides the classical reference model for gravitational interactions in the weak-field limit. In this work, it is used as the baseline model against which the relativistic correction is quantified.

The Newtonian approximation assumes a fixed space-time framework and is accurate when gravitational fields are sufficiently weak. However, near strong-field regions, deviations from the Newtonian description become significant, motivating comparison with a relativistic formulation.

The mathematical definition and normalization of the Newtonian potential used in this study are presented in Section 4.2.

3.4 Static Relativistic Potential Model

To incorporate strong-field gravitational effects, this study considers a static relativistic potential derived from the Schwarzschild solution of general relativity.

Unlike the Newtonian approximation, the relativistic model introduces nonlinear corrections arising from spacetime curvature. These corrections become increasingly important near the Schwarzschild radius.

The complete mathematical formulation, normalization procedure, and dimensionless representation of the relativistic potential are described in Section 4.1.

3.5 Relativistic-Newtonian Comparison

The Newtonian and relativistic potential models describe the same gravitational system but differ in their treatment of gravitational effects.

The Newtonian potential provides a classical approximation that assumes weak gravitational fields and neglects relativistic corrections. In contrast, the relativistic potential incorporates nonlinear effects that arise from the relativistic description of gravity.

A comparison of the two models allows the accuracy of the classical approximation to be evaluated as a function of the normalized radial coordinate. In

regions far from the characteristic relativistic scale, the two potentials exhibit similar functional behavior, allowing Newtonian gravity to provide a qualitative approximation. However, quantitative comparison through relative error analysis is required to determine the accuracy of this approximation.

Closer to the strong-field region, however, the difference between the models becomes increasingly significant. This deviation reflects the growing importance of relativistic corrections and motivates the use of quantitative error analysis.

To measure the discrepancy between the two descriptions, a relative error function is introduced in the following section. This function provides a numerical basis for evaluating the validity range of the Newtonian approximation and identifying the transition between classical and relativistic gravitational behavior.

4 Mathematical Formulation

4.1 Relativistic Potential Equation

The relativistic gravitational potential investigated in this study is derived from the Schwarzschild solution and is expressed as

$$\Phi_{GR}(r) = \frac{c^2}{2} \left(\sqrt{1 - \frac{2GM}{rc^2}} - 1 \right).$$

where:

- G represents the gravitational constant.
- M represents the mass of the gravitating source.
- r represents the radial distance.
- c represents the speed of light.

Introducing the Schwarzschild radius

$$r_s = \frac{2GM}{c^2}$$

and the dimensionless coordinate about which we discussed in [Section 3.2](#).

$$x = \frac{r}{r_s},$$

Substituting:

$$GM = \frac{r_s c^2}{2}$$

and

$$r = x r_s,$$

we obtain:

$$\Phi_{GR}(x) = \frac{c^2}{2} \left(\sqrt{1 - \frac{1}{x}} - 1 \right).$$

the dimensionless Relativistic potential is obtained by normalizing with c^2 :

$$\phi_{GR}(x) = \frac{\Phi_{GR}}{c^2} = \frac{1}{2} \left(\sqrt{1 - \frac{1}{x}} - 1 \right)$$

This equation represents the reference relativistic model used throughout the study. The nonlinear square-root dependence introduces deviations from classical gravity that become significant near the Schwarzschild radius.

For large values of (x) , the relativistic potential follows the same inverse-distance dependence as the Newtonian approximation. However, because the leading-order coefficients differ, the two potentials do not converge to the same normalized value. The difference arises because the adopted relativistic potential is normalized through the Schwarzschild metric expression rather than constructed as the classical Newtonian limit; therefore the comparison is performed within the defined potential formulation. This difference is quantified through the relative error analysis performed in later sections. Therefore, the weak-field limit in this formulation should be interpreted as convergence in functional behavior rather than exact numerical agreement between the two normalized potentials.

4.2 Newtonian Potential Equation

Newtonian Potential Equation The Newtonian gravitational potential is taken as the classical reference potential for comparison purposes with the relativistic potential equation. The potential equation represents the gravitational effect of the mass source assuming that the effects of spacetime and the gravitational field are in the weak-field approximation region.

The standard Newtonian gravitational potential is given by:

$$\Phi_N(r) = -\frac{GM}{r}$$

where:

- G represents the gravitational constant.
- M represents the mass of the gravitational source.
- r represents the radial distance from the source.

Using the dimensionless radial coordinate introduced previously in Section 3.2:

$$x = \frac{r}{r_s}$$

and the Schwarzschild radius

$$r_s = \frac{2GM}{c^2}$$

the normalized Newtonian potential becomes:

$$\phi_N(x) = \frac{\Phi_N}{c^2} = -\frac{GM}{rc^2}$$

Substituting:

$$GM = \frac{r_s c^2}{2}$$

and

$$r = xr_s,$$

we obtain:

$$\phi_N(x) = -\frac{1}{2x}$$

This normalized expression provides a simplified representation that allows direct comparison with the relativistic potential model.

The Newtonian approximation serves as the standard for which the relativistic corrections are taken into consideration. For large values of (x), both models exhibit similar inverse-distance behavior. However, the relative error analysis is required to determine the actual quantitative agreement between the two descriptions. However, as (x) goes deeper into stronger-field regions, the relativistic potential starts to differ from the classical approximation.

4.3 Derivation of Relative Error Expression

To quantify the difference between the relativistic and Newtonian descriptions, the relative error between the two potential models is calculated.

The relative error provides a measure of how much the Newtonian approximation deviates from the relativistic potential at a given value of the normalized radial coordinate (x).

The general form of the relative error is defined as:

$$\epsilon(x) = \left| \frac{\phi_{GR}(x) - \phi_N(x)}{\phi_{GR}(x)} \right| \times 100$$

where:

- $\phi_{GR}(x)$ represents the normalized relativistic potential,
- $\phi_N(x)$ represents the normalized Newtonian approximation,
- $\epsilon(x)$ represents the percentage deviation between the two models.

Substituting the expressions for the relativistic and Newtonian potentials into the error equation gives a mathematical relationship describing the accuracy of the classical approximation across different gravitational regimes.

This error function allows the transition between relativistic and classical behavior to be studied quantitatively. When $\epsilon(x)$ is small, the Newtonian model provides a close approximation to the relativistic system. However, an increasing value of $\epsilon(x)$ indicates that relativistic corrections become significant.

The resulting error function is evaluated computationally over a range of (x) values to analyze the behavior of the approximation and identify regions where classical gravity remains sufficiently accurate.

This formulation provides the connection between the theoretical model and the numerical analysis performed in this study.

4.4 Simplified Error Formula

After substituting the relativistic and Newtonian potential expressions into the relative error equation, the resulting expression can be simplified to obtain a direct relationship between the error and the normalized radial coordinate (x) .

The simplified form obtained is:

$$\epsilon(x) = 100 \left| \frac{-x^{3/2} + \sqrt{x} + x\sqrt{x-1}}{x(-\sqrt{x} + \sqrt{x-1})} \right|$$

This expression represents the percentage deviation between the relativistic potential and the Newtonian approximation as a function of the dimensionless radial coordinate.

The equation provides a direct way to analyze how the accuracy of the Newtonian model changes across different gravitational regimes. As (x) increases, the relativistic corrections become smaller and the error approaches the weak-field limit, where the Newtonian approximation becomes increasingly accurate.

Conversely, for values of (x) closer to the strong-field region, the deviation between the two models becomes more significant. This behavior reflects the increasing influence of relativistic effects as the system moves away from the conditions where classical gravity provides a reliable approximation.

The simplified error expression is particularly useful because it removes the need to repeatedly compare the two potential functions separately. Instead, the accuracy of the Newtonian approximation can be directly evaluated from a single mathematical relationship.

This analytical formulation is later used for numerical error analysis, including the determination of accuracy thresholds and the identification of the transition region between relativistic and classical gravitational behavior.

5 Error Analysis

5.1 Error Behaviour

The relative error function derived in the previous section provides a quantitative measure of the difference between the relativistic potential and the Newtonian approximation. By evaluating this function over a range of values of the normalized radial coordinate (x), the accuracy of the classical model can be studied.

The behavior of the error is expected to depend strongly on the gravitational regime being considered. In regions closer to the characteristic relativistic scale, the effects of relativistic corrections become more significant, resulting in a larger deviation between the two potential models.

As the value of (x) increases, the gravitational field becomes weaker and the functional behavior of the relativistic and Newtonian descriptions becomes increasingly similar. However, due to the difference in their asymptotic normalization, the relative error does not decrease to zero and instead approaches a limiting value.

The variation of the relative error with respect to the normalized radial coordinate is shown in Figure 1.

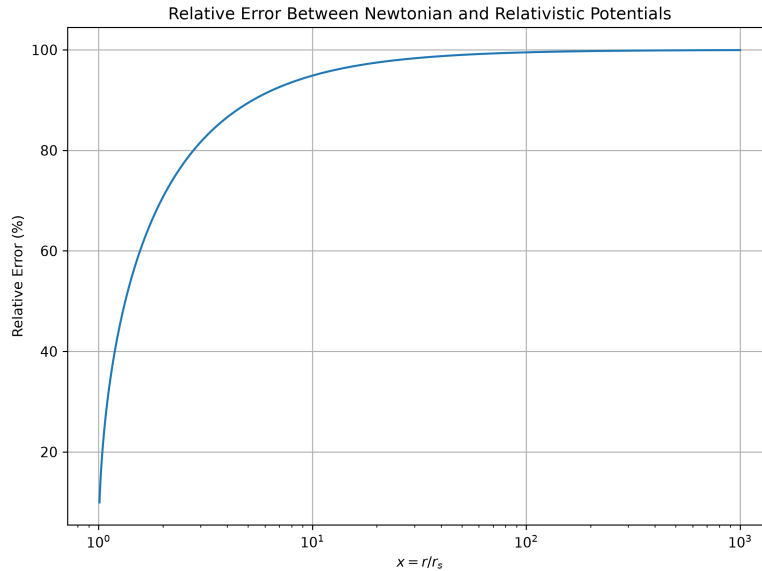


Figure 1: Relative error between the normalized relativistic potential and the Newtonian approximation as a function of the dimensionless radial coordinate x . The minimum error occurs near $x = 1.01$, while the error increases toward the weak-field regime due to the differing asymptotic normalizations of the two models.

This behavior demonstrates the transition from a region where relativistic

corrections significantly modify the potential toward a region where both models exhibit similar qualitative inverse-distance behavior. However, due to the difference in asymptotic normalization, the Newtonian model does not achieve vanishing relative error under the chosen formulation.

The numerical evaluation of the error function allows this transition to be examined quantitatively rather than only through theoretical assumptions. By measuring how the error changes with (x) , the range in which Newtonian gravity achieves a specified accuracy can be determined.

The error analysis therefore provides an important link between the theoretical formulation and the quantitative evaluation of the classical approximation within the considered model.

5.2 Error Threshold Analysis

A numerical threshold analysis was performed to determine whether the Newtonian approximation reaches commonly used accuracy levels within the sampled domain.

The threshold condition was defined as:

$$\epsilon(x) \leq \epsilon_{limit}$$

where ϵ_{limit} represents a specified maximum relative error.

Three thresholds were investigated:

- 10% relative error,
- 5% relative error,
- 1% relative error.

The numerical evaluation showed that the relative error reaches its minimum value near the lower boundary of the investigated domain:

$$x = 1.01$$

with a corresponding relative error of approximately:

$$\epsilon = 9.950372$$

However, the analysis further revealed that the relative error never falls below either the 5% or 1% thresholds within the investigated domain:

$$1.01 \leq x \leq 1000$$

The minimum observed relative error across the sampled dataset was approximately 9.95%.

These results indicate that, for the specific relativistic potential considered in this study, the Newtonian approximation does not achieve better than approximately 10% accuracy within the investigated domain. The minimum relative error occurs near the lower boundary of the sampled range ($x = 1.01$), while the discrepancy increases as x becomes larger.

This behavior arises from the different asymptotic forms of the two potentials. In the large-distance limit, the relativistic potential approaches $\phi_{GR} \approx -1/(4x)$, whereas the Newtonian approximation follows $\phi_N = -1/(2x)$. Consequently, the relative error increases toward 100% as $x \rightarrow \infty$ rather than remaining near 10%. Therefore, the observed 9.95% value represents the minimum error within the chosen domain, not the asymptotic error behavior.

5.3 Strong-Field and Weak-Field Transition

The variation of the relative error with the normalized radial coordinate provides a way to identify the transition between strong-field and weak-field gravitational regimes.

In the strong-field region, where (x) approaches the characteristic relativistic scale, the assumptions behind the Newtonian approximation become less accurate. The contribution of relativistic corrections becomes more significant, leading to a larger difference between the relativistic and classical potential models.

As the value of (x) increases, the influence of relativistic corrections gradually decreases. The system moves toward the weak-field regime, where the Newtonian approximation becomes increasingly consistent with the relativistic description.

This transition can be understood through the behavior of the relative error. A larger error indicates that the classical approximation does not fully capture the physical behavior of the system, while a smaller error indicates that the Newtonian model provides an effective approximation.

By analyzing the points where the error crosses predefined accuracy limits, the boundary between these regimes can be estimated quantitatively. Instead of defining the transition only through theoretical assumptions, this approach provides a measurable criterion based on the accuracy of the approximation.

The identification of this transition region is important because it establishes the range in which classical gravitational models remain applicable and the range where relativistic corrections become necessary. This analysis also provides a physically meaningful interpretation of the dataset later used for machine learning experiments.

5.4 Asymptotic Behaviour

The asymptotic behavior of the relativistic and Newtonian potentials provides important insight into their relationship in the weak-field regime.

As the normalized radial coordinate (x) increases, both potentials approach an inverse-distance dependence, indicating similar qualitative behavior. However, they do not converge to the same value because their leading-order terms differ:

$$\phi_{GR} \approx -\frac{1}{4x}$$

whereas

$$\phi_N = -\frac{1}{2x}.$$

Therefore, the Newtonian approximation captures the general distance-dependent trend of the relativistic potential but retains a systematic quantitative discrepancy due to the difference in normalization. The observed asymptotic discrepancy is a consequence of the chosen normalization of the relativistic potential and should not be interpreted as a failure of the Newtonian weak-field approximation itself.

Numerical evaluation of the relative error reveals that the minimum observed deviation within the investigated domain occurs near the lower boundary:

$$\epsilon_{min} = 9.950372\%$$

at approximately:

$$x = 1.01.$$

The difference between the relativistic and Newtonian potentials is shown in Figure 2, illustrating the systematic deviation between the two models.

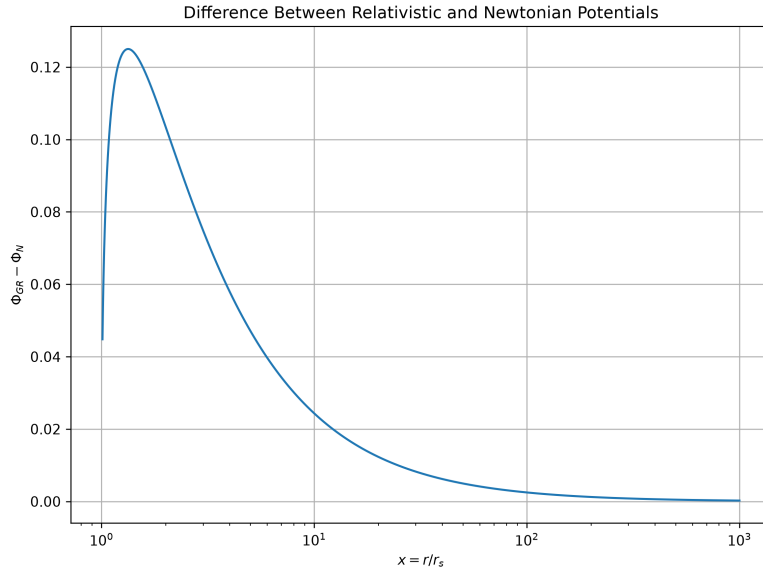


Figure 2: Difference $\Delta\phi = \phi_{GR} - \phi_N$ between the normalized relativistic and Newtonian potential models across the investigated domain. The plot illustrates the systematic deviation between the two formulations.

However, this minimum value should not be interpreted as the asymptotic error. As x increases, the relative error grows and approaches 100% in the large-distance limit because the ratio between the leading-order terms remains different.

This behavior explains why the 5% and 1% accuracy thresholds are never reached within the sampled range:

$$1.01 \leq x \leq 1000.$$

The location of the maximum deviation is further illustrated in Figure 3.

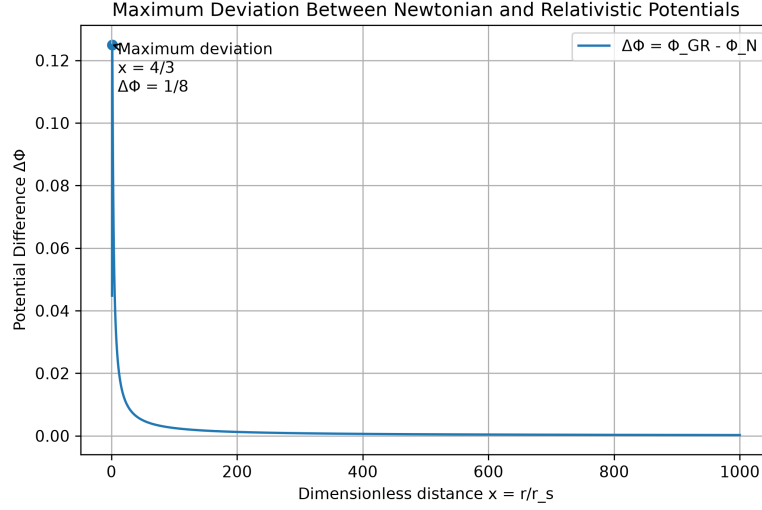


Figure 3: Annotated plot of the potential difference showing the location of maximum deviation between the relativistic and Newtonian models.

Therefore, while the Newtonian approximation reproduces the qualitative inverse-distance behavior of the relativistic potential, explicit error analysis reveals a persistent and increasing quantitative discrepancy. This result demonstrates the importance of evaluating approximation accuracy through numerical comparison rather than assuming convergence solely from similar functional behavior.

6 Gravitational Field Analysis

6.1 Newtonian Field

The gravitational field is obtained from the spatial variation of the potential. In the Newtonian framework, the gravitational field is defined as the negative gradient of the gravitational potential:

$$g_N = -\frac{d\Phi_N}{dr}$$

Using the Newtonian potential:

$$\Phi_N(r) = -\frac{GM}{r}$$

the corresponding gravitational field becomes:

$$g_N = \frac{GM}{r^2}$$

This represents the classical inverse-square behavior of gravity, where the field strength decreases with increasing distance from the source.

Using the normalized coordinate (x), the Newtonian field can be analyzed in terms of the relative distance scale rather than absolute units. This allows direct comparison with the relativistic field representation.

The Newtonian field provides the classical reference behavior for evaluating the influence of relativistic corrections. In the weak-field regime, it is expected to closely match the relativistic prediction, while deviations become more noticeable as the system approaches stronger gravitational regions.

The analysis of the Newtonian field therefore provides the baseline required for understanding how relativistic effects modify the gravitational behavior of the system.

6.2 Relativistic Field

In the relativistic description, the gravitational field is obtained from the variation of the relativistic potential with respect to the radial coordinate:

$$g_{GR} = -\frac{d\Phi_{GR}}{dr}$$

Unlike the Newtonian formulation, the relativistic field includes corrections that arise from the modified structure of the gravitational potential. These corrections become increasingly important in regions where the gravitational field is stronger.

The behavior of the relativistic field differs from the Newtonian prediction mainly due to the nonlinear dependence introduced by the relativistic formulation. At larger values of (x), where the gravitational field is weaker, the relativistic field gradually approaches the Newtonian result. However, closer to the strong-field region, the difference between the two descriptions becomes more significant.

The comparison of these two field models provides additional information beyond the potential analysis. While the potential comparison measures the difference in gravitational energy representation, the field comparison examines how the gravitational influence itself changes across different regimes.

This analysis helps establish a more complete understanding of the system by considering both the potential and the resulting field behavior before applying computational and machine learning methods.

6.3 Comparative Analysis

The comparison between the Newtonian and relativistic gravitational fields provides a broader perspective on the transition between classical and relativistic descriptions.

In the weak-field regime, the difference between the two models becomes minimal, indicating that Newtonian gravity provides an accurate approximation of the relativistic behavior. This agreement explains why classical gravitational models remain effective for many practical applications where relativistic corrections are extremely small.

However, as the system moves toward stronger gravitational regions, the relativistic corrections become increasingly important. The deviation between the two field descriptions reflects the limitations of the classical approximation and highlights the need for a relativistic treatment.

The combined analysis of potential and field behavior demonstrates that the transition from Newtonian to relativistic behavior is gradual rather than abrupt. The accuracy of the classical approximation depends on the physical regime being considered and can be quantified through computational error analysis.

This comparison provides additional validation for the dataset generation process used in this study. Since the machine learning models are trained on values derived from the relativistic formulation, understanding the physical behavior of the underlying system is essential for interpreting the predictions produced by the models.

7 Dataset Construction

7.1 Data Generation

The dataset used in this study was generated directly from the mathematical formulation of the static relativistic potential model. Instead of relying on externally collected data, the values were produced by evaluating the theoretical equations across a defined range of the normalized radial coordinate (x).

For each value of (x), the corresponding relativistic potential value was calculated and stored as the target physical quantity for machine learning analysis. The Newtonian approximation and error-related quantities were also evaluated to support comparison and validation.

The generated dataset represents the relationship between the input parameter and the resulting gravitational potential behavior. This approach ensures that the machine learning models are trained on data that follows the underlying physical laws represented by the mathematical model.

The dataset generation process can be summarized as:

1. Selecting a range of normalized radial coordinate values (x).
2. Evaluating the relativistic potential model for each value.
3. Storing the resulting input-output relationship.
4. Using the generated dataset for training and evaluating machine learning models.

By generating the dataset from the analytical model, the study creates a controlled environment where machine learning performance can be evaluated against a known physical relationship.

7.2 Dataset Range and Sampling

To analyze the behavior of the relativistic potential across different gravitational regimes, the dataset was generated over multiple ranges of the normalized radial coordinate (x).

Two different sampling ranges were considered in this study:

1. A focused dataset covering a smaller range of (x), designed to capture the behavior closer to the stronger-field region where relativistic effects are more significant.
2. A broader dataset extending to larger values of (x), allowing the analysis of the weak-field regime where the relativistic potential approaches the Newtonian approximation.

The smaller-range dataset was used for detailed analysis of the region where the potential changes more rapidly, while the extended-range dataset provided a wider representation of the overall behavior of the system.

Sampling across these ranges allows the machine learning models to learn both the nonlinear behavior present in stronger gravitational regions and the gradual convergence toward classical behavior at larger distances.

The choice of sampling range is important because machine learning models learn patterns from the provided data distribution. A dataset that includes both strong-field and weak-field behavior allows the trained models to develop a more complete representation of the underlying physical relationship.

This approach ensures that the computational experiments evaluate not only the predictive capability of machine learning models but also their ability to approximate physical behavior across different regimes.

7.3 Statistical Properties

Before applying machine learning algorithms, the generated dataset was analyzed to understand its structure and characteristics. Since the dataset is derived from a mathematical model, its statistical properties directly reflect the behavior of the underlying physical system.

The primary input variable is the normalized radial coordinate:

$$x = \frac{r}{r_s}$$

which represents the position parameter of the system. The corresponding target variable is the relativistic potential:

$$\phi_{GR}(x)$$

which the machine learning models are trained to approximate.

The relationship between these variables is nonlinear, particularly in regions where relativistic effects become significant. This nonlinear behavior provides a suitable test case for evaluating different machine learning approaches, ranging from simple regression models to more flexible nonlinear models.

The dataset characteristics are important because the performance of a machine learning model depends strongly on the complexity and distribution of the available data. Regions with rapid changes in the potential require models capable of capturing nonlinear patterns, while smoother regions provide a simpler approximation problem.

The generated dataset therefore serves as a controlled scientific benchmark where the accuracy of machine learning models can be evaluated against a known physical relationship.

The combination of theoretical generation and statistical analysis ensures that the machine learning results can be interpreted in the context of the original gravitational model rather than as purely data-driven predictions.

The generated dataset contains 10,000 samples spanning the normalized radial coordinate range from $x = 1.01$ to $x = 1000$. Descriptive statistical analysis was performed to characterize the dataset prior to machine learning training.

The mean value of the normalized radial coordinate was 144.86, while the mean relativistic potential was -0.04379 . The relative error between the Newtonian and relativistic models ranged from approximately 9.95% to 99.95%, with an average value of 91.24%.

These statistics indicate that significant deviations exist between the classical and relativistic descriptions throughout much of the sampled range. The wide variation of the potential values and error measurements provides a suitable benchmark for evaluating the approximation capabilities of different machine learning models.

8 Machine Learning Framework

8.1 Dataset Generation and Preparation

The numerical analysis and machine learning experiments were conducted using datasets generated specifically for their respective objectives.

For analytical investigations, including error evaluation, threshold analysis, and asymptotic behaviour analysis, a dataset covering the range:

$$1.01 \leq x \leq 1000$$

was generated. This extended range enables investigation of both strong-field and weak-field gravitational regimes.

For machine learning experiments, a restricted dataset:

$$1.01 \leq x \leq 100$$

was considered. At significantly larger values of x , the relativistic potential approaches a slowly varying weak-field limit, resulting in reduced variation

in the target function. This decreases the information content available for model training. Therefore, the restricted range was selected to provide a more representative description of the nonlinear behaviour present within the potential.

The machine learning models were trained and evaluated using an 80:20 train-test split. The dataset consists of the normalized radial coordinate x as the input feature and the relativistic potential

$$\phi_{GR}(x)$$

as the target variable. Additional physical quantities, including the Newtonian potential, potential difference, and relative error, were retained for numerical analysis but were excluded from the machine learning input features.

8.2 Problem Definition

The objective of the machine learning component is to develop computational models capable of approximating the behaviour of the static relativistic potential using data generated from the analytical formulation.

The problem is formulated as a supervised learning task, where the normalized radial coordinate acts as the input variable and the corresponding relativistic potential represents the target output:

$$x \rightarrow \phi_{GR}(x)$$

The purpose of the machine learning models is to learn an efficient approximation of the relationship between the input parameter and the physical quantity, allowing rapid evaluation without repeatedly calculating the original analytical expression.

Multiple machine learning approaches with different levels of complexity are investigated to analyse how model architecture influences approximation capability. The selected models represent different learning strategies, including traditional regression, nonlinear function approximation, and ensemble-based learning.

The evaluated models include:

- Linear regression, which provides a baseline approximation for the relativistic potential.
- Polynomial regression, which introduces nonlinear feature expansion to improve representation capability.
- Neural networks, which can learn complex nonlinear relationships from data.
- Random forest regression, which applies ensemble learning to capture nonlinear dependencies within the dataset.

Each model is evaluated using standard regression metrics and compared against the analytical solution to determine its ability to reproduce the relativistic potential.

This framework enables investigation of both the accuracy of individual machine learning approaches and the influence of different model architectures when applied to a physically generated dataset.

All machine learning models were implemented using the Scikit-learn library [6].

8.3 Linear Regression

Linear regression is considered as the initial baseline model to evaluate whether a simple mathematical approximation can represent the behaviour of the relativistic potential.

The model assumes a linear relationship between the input variable and the target output, expressed as:

$$\hat{y} = ax + b$$

where:

- $\hat{y}$ represents the predicted potential value.
- x represents the normalized radial coordinate.
- a and b represent the parameters optimized during training.

Although the actual relationship between x and $\phi_{GR}(x)$ is nonlinear, linear regression provides a useful reference point for evaluating the inherent complexity of the approximation problem.

The purpose of including this model is not to obtain maximum prediction accuracy, but to establish a quantitative baseline. Poor performance of a linear model indicates that the underlying physical relationship contains nonlinear characteristics that cannot be represented using a simple first-order approximation.

The trained linear regression model is evaluated using standard regression metrics and compared with more advanced models in subsequent sections.

This comparison demonstrates how increasing model flexibility influences the capability of machine learning approaches to approximate the relativistic potential.

Regression-based approaches are commonly used as baseline methods in statistical learning and predictive modelling applications [4, 5].

8.4 Polynomial Regression

Polynomial regression is introduced to examine whether incorporating higher-order nonlinear terms improves approximation of the relativistic potential compared with a linear model.

Unlike linear regression, polynomial regression expands the input variable into higher-order terms:

$$\hat{y} = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$

where the coefficients $a_0, a_1, \dots, a_n$ are learned from the training data.

By introducing additional powers of x , polynomial regression gains the ability to represent curved relationships and approximate nonlinear functions. Therefore, it provides an intermediate approach between simple linear models and more flexible machine learning architectures.

However, increasing polynomial order can introduce overfitting, where the model captures specific characteristics of the training dataset but fails to generalize effectively to unseen data.

In this study, polynomial regression is used to investigate whether mathematical feature expansion alone is sufficient to capture the nonlinear structure of the relativistic potential. Its performance is compared with both simpler regression methods and higher-capacity machine learning models.

The results provide insight into whether the complexity of the physical relationship can be represented through analytical transformations or requires more flexible learning architectures.

8.5 Neural Network

A neural network model is introduced to investigate whether a flexible learning architecture can effectively approximate the nonlinear relationship present in the relativistic potential system.

Unlike traditional regression techniques, neural networks learn complex relationships through interconnected computational units arranged in multiple layers. The theoretical foundation of neural networks and deep learning approaches is described in modern machine learning literature [7]. A typical neural network consists of an input layer, one or more hidden layers, and an output layer.

In this study, the input variable is the normalized radial coordinate:

$$x$$

and the predicted output is the relativistic potential:

$$\hat{\phi}_{GR}(x)$$

The network learns an approximation function:

$$f(x) \approx \phi_{GR}(x)$$

during training by adjusting internal parameters to minimize the difference between predicted and analytical values.

The primary advantage of neural networks in this context is their ability to represent nonlinear relationships without requiring an explicit functional form. Since the relativistic potential exhibits nonlinear behaviour, this flexibility allows the model to capture patterns that simpler regression methods may not represent accurately.

The neural network performance is evaluated using regression metrics and compared with other models. The objective is not to replace the analytical

formulation, but to examine whether a learned approximation can reproduce the behaviour of the physical model with sufficient accuracy.

The resulting analysis provides a comparison between analytical modelling and data-driven approximation methods for a nonlinear scientific system.

8.6 Random Forest

Random forest regression is included as an ensemble-based machine learning approach for approximating the relativistic potential.

A random forest consists of multiple decision trees trained on different subsets of the dataset. The approach follows the ensemble learning methodology introduced by Breiman [8]. Each individual tree learns decision-based relationships between the input variable and target output, and the final prediction is obtained by combining the predictions from all trees.

The prediction process can be represented as:

$$\hat{y} = \frac{1}{N} \sum_{i=1}^N T_i(x)$$

where:

- $T_i(x)$ represents the prediction from the i^{th} decision tree.
- N represents the total number of trees.
- $\hat{y}$ represents the final predicted output.

The advantage of random forest regression is its ability to model nonlinear relationships while reducing sensitivity to individual tree variations. Since the relativistic potential exhibits nonlinear behaviour, ensemble-based learning provides an effective method for capturing complex patterns within the generated dataset.

In this study, random forest regression is evaluated alongside regression-based approaches and neural networks to compare different learning strategies.

The high predictive accuracy obtained by the model demonstrates that machine learning techniques can successfully approximate the numerical behaviour of the relativistic potential within the investigated dataset range. However, this performance should be interpreted in the context of a deterministic analytical dataset. Since the training data originates from the known physical model, the machine learning model reproduces an existing relationship rather than independently discovering the underlying physical law.

8.7 Feature Importance Analysis

In addition to evaluating prediction accuracy, understanding the relationship between input variables and model predictions is an important aspect of interpreting machine learning systems.

Feature importance analysis is included as an interpretability step to verify that the machine learning model relies on the physically relevant input parameter used in the formulation. In this study, the primary feature considered is the normalized radial coordinate:

$$x = \frac{r}{r_s}$$

Since the dataset is generated from a single physical variable, the analysis focuses on understanding the influence of changes in x on the predicted relativistic potential.

For tree-based models such as random forest regression, feature importance is determined by analysing the contribution of the feature in reducing prediction uncertainty during model construction.

This analysis connects the machine learning predictions with the underlying physical formulation. Rather than treating the model as a complete black box, it provides an additional verification that the learned relationship remains consistent with the expected dependence of the potential on radial distance.

Although the present system contains only one dominant input parameter, this analysis establishes a foundation for future extensions involving multiple physical variables, where interpretability becomes increasingly important. The feature importance analysis is included as an interpretability assessment rather than as a comparison between multiple physical variables. Since the present formulation contains a single independent parameter, the analysis confirms that the machine learning model preserves the expected dependence of the relativistic potential on the normalized radial coordinate.

The combination of prediction performance and interpretability analysis allows the machine learning results to be evaluated from both computational and scientific perspectives.

9 Results and Analysis

9.1 Model Performance Comparison

The predictive capability of the machine learning models was evaluated using three widely adopted regression performance metrics: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2) [4, 5]. These metrics collectively quantify prediction accuracy, sensitivity to large errors, and the proportion of variance explained by each model.

The resulting performance measures are summarized in Table 1.

Table 1: Performance comparison of machine learning models.

Model	MAE	RMSE	R^2
Linear Regression	0.05579	0.07709	0.26914
Polynomial Regression	0.05624	0.07908	0.23091
Neural Network	0.00745	0.01038	0.98347
Random Forest	0.000028	0.000084	0.999999

The corresponding predictions produced by the machine learning models are compared with the analytical relativistic potential in Figure 4.

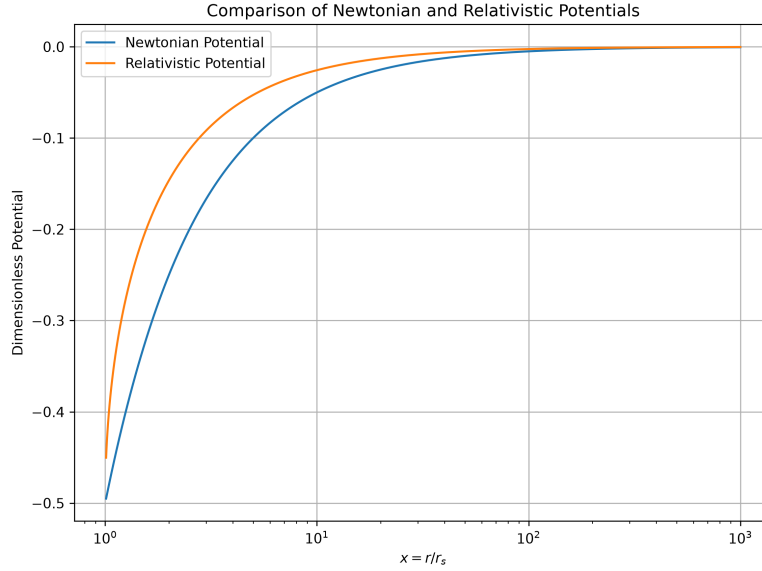


Figure 4: Comparison of machine learning predictions with the analytical relativistic potential. The figure illustrates the ability of different models to reproduce the theoretical relationship.

A clear hierarchy in predictive performance is observed among the evaluated algorithms. Linear regression and polynomial regression exhibit comparatively large prediction errors, indicating that the dependence of the relativistic potential on the normalized radial coordinate is inherently nonlinear and cannot be represented adequately using low-order regression models.

The neural network substantially improves the approximation accuracy, achieving an R^2 value exceeding 0.98 while simultaneously reducing both the MAE and RMSE. This demonstrates its ability to learn the nonlinear structure of the generated physical dataset with high fidelity.

Among all evaluated models, random forest regression provides the highest predictive accuracy, yielding an almost perfect coefficient of determination

($R^2 \approx 0.999999$) together with extremely small prediction errors. The ensemble-based learning strategy enables the model to reproduce the analytical relativistic potential with remarkable precision throughout the investigated machine-learning domain. Such exceptional performance is expected because the training dataset is generated directly from a deterministic analytical function involving a single independent variable.

Overall, the results demonstrate that nonlinear machine learning methods provide substantially more accurate approximations of the relativistic potential than conventional regression techniques. They also illustrate the importance of selecting learning architectures capable of representing nonlinear physical relationships.

9.1.1 Numerical Error Analysis Summary

In addition to evaluating machine learning performance, the analytical formulation was investigated through numerical error analysis. The difference between the relativistic and Newtonian potentials is defined as

$$\Delta\phi = \phi_{GR} - \phi_N$$

The principal numerical results obtained from the computational analysis are summarized in Table 2.

Table 2: Summary of numerical error analysis results.

Quantity	Value
Number of samples	10,000
Numerical Analysis Domain	$1.01 \leq x \leq 100$
Maximum deviation location	$x = \frac{4}{3}$
Maximum deviation	$\Delta\phi_{\max} = \frac{1}{8}$
Minimum relative error	9.950372%
Maximum relative error	99.949987% (approaching 100% as $x \rightarrow \infty$)

The numerical investigation reveals that the minimum relative error occurs at the lower boundary of the computational domain ($x = 1.01$), where the discrepancy between the relativistic and Newtonian potentials is approximately 9.95%. Throughout the investigated interval, neither the 5% nor the 1% relative-error thresholds are attained.

As the normalized radial coordinate increases, the relative error rises monotonically and approaches approximately 100%. This asymptotic behaviour originates from the fundamentally different mathematical forms of the relativistic and Newtonian potential functions. These results quantitatively illustrate the limitations

of the Newtonian approximation within the adopted dimensionless formulation and provide a computational characterization of the transition between relativistic and classical gravitational descriptions.

9.2 Prediction Analysis

To further assess the predictive capability of the evaluated machine learning algorithms, the model outputs were compared directly with the analytical relativistic potential generated from the theoretical formulation. Such comparisons provide a visual assessment of how accurately each learning algorithm reproduces the underlying physical relationship in addition to the numerical performance metrics presented previously.

The prediction curves illustrate the agreement between the normalized radial coordinate (x) and the corresponding relativistic potential $\phi_{GR}(x)$ across the investigated machine-learning domain.

The linear and polynomial regression models exhibit noticeable deviations from the analytical solution, particularly in regions where the potential varies nonlinearly. This behaviour reflects the limited representational capacity of these relatively simple regression techniques when applied to nonlinear physical systems.

The neural network produces a substantially improved approximation, closely following the analytical solution over the entire computational domain. Its ability to learn complex nonlinear mappings enables the model to reproduce the variation of the relativistic potential with considerably greater accuracy than conventional regression approaches.

Among the evaluated models, random forest regression provides the closest agreement with the analytical solution. The predicted values remain nearly indistinguishable from the theoretical curve throughout the investigated range, demonstrating that ensemble learning methods can effectively approximate deterministic nonlinear physical relationships when trained using sufficiently representative datasets.

The prediction capability of the random forest model is illustrated in Figure 5.

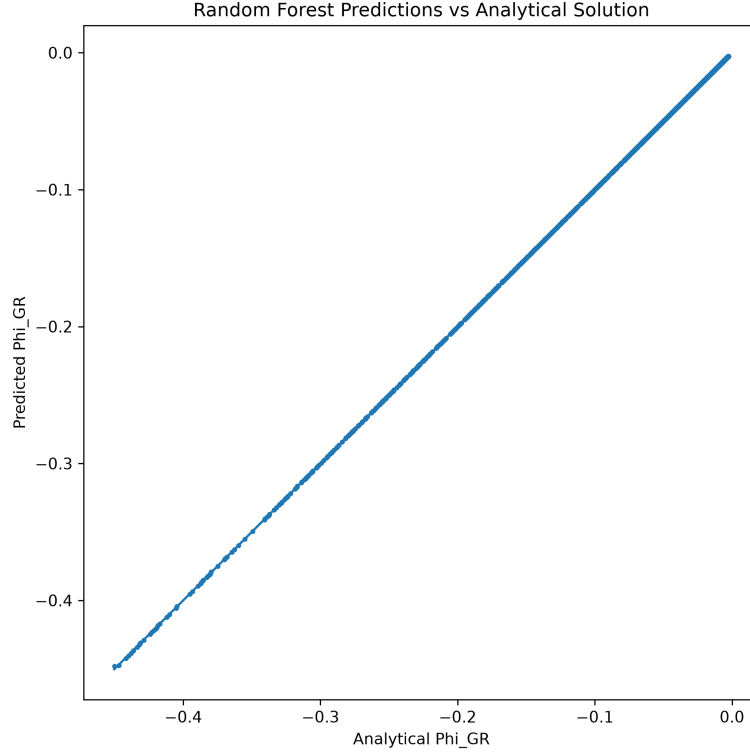


Figure 5: Random forest regression predictions compared with the analytical relativistic potential over the machine-learning dataset range.

The prediction analysis supports the conclusions obtained from the quantitative performance metrics. Machine learning models possessing greater nonlinear representational capacity consistently provide more accurate approximations of the relativistic potential than simpler regression-based methods.

It should be emphasized, however, that the objective of the present investigation is not to replace the underlying physical model with a purely data-driven alternative. Since the training data are generated directly from the analytical relativistic potential, the machine learning algorithms function as computational surrogate models rather than independent methods for discovering physical laws. Their performance therefore reflects the ability to reproduce an established analytical relationship efficiently while preserving the essential characteristics of the underlying gravitational model.

9.3 Residual Analysis

To obtain a more detailed assessment of prediction accuracy, the performance of the machine learning models was further examined through residual analysis.

Unlike global performance metrics such as MAE and RMSE, residual analysis provides information about the distribution of prediction errors throughout the computational domain and helps identify systematic deviations between the analytical model and its machine learning approximation.

The residual is defined as the difference between the analytical relativistic potential and the corresponding machine learning prediction:

$$\text{Residual} = \phi_{GR}(x) - \hat{\phi}_{GR}(x)$$

where:

- $\phi_{GR}(x)$ represents the analytical relativistic potential.
- $\hat{\phi}_{GR}(x)$ represents the machine learning prediction.

Residual analysis complements conventional regression metrics by revealing how prediction errors are distributed over the input domain. A model exhibiting small, randomly distributed residuals generally indicates that the underlying physical relationship has been captured successfully, whereas systematic residual patterns may suggest deficiencies in the model structure or approximation capability.

The linear and polynomial regression models exhibit comparatively larger residuals, particularly in regions where the relativistic potential varies nonlinearly. These deviations are consistent with the limited flexibility of simple regression models and further demonstrate that low-order mathematical approximations are insufficient for accurately representing the nonlinear behaviour of the relativistic potential.

In contrast, the neural network and random forest models produce substantially smaller residuals across the investigated domain. The reduction in prediction error confirms their ability to capture the nonlinear dependence of the relativistic potential on the normalized radial coordinate with significantly greater accuracy.

Residual analysis also provides insight into the local reliability of machine learning predictions. Regions exhibiting comparatively larger residuals identify portions of the parameter space where approximation becomes more challenging and therefore provide useful information regarding the limitations of each learning algorithm.

Consequently, residual analysis complements the global performance metrics presented previously by providing a more comprehensive evaluation of how effectively each machine learning model reproduces the analytical relativistic potential.

The residual distribution obtained from the machine learning predictions is illustrated in Figure 6.

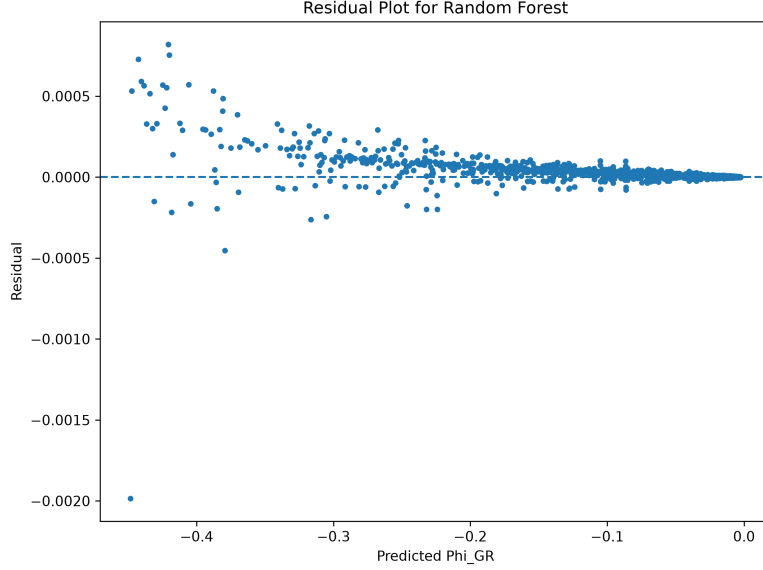


Figure 6: Residual distribution obtained from machine learning predictions, showing the difference between analytical and predicted relativistic potential values.

9.4 Feature Importance Results

In addition to predictive accuracy, the interpretability of machine learning models is an important consideration in scientific applications. Feature importance analysis was therefore performed to examine the contribution of the input variable to the prediction process and to verify that the learned relationship remains consistent with the underlying physical formulation.

The present dataset contains a single physical input parameter,

$$x = \frac{r}{r_s},$$

which represents the normalized radial coordinate. Since the relativistic potential is explicitly defined as a function of this variable, the prediction produced by each machine learning model is entirely determined by the variation of x .

For tree-based algorithms such as random forest regression, feature importance is evaluated by measuring the contribution of each input variable to the reduction of prediction uncertainty throughout the ensemble of decision trees. Although the present study involves only a single input feature, this analysis provides an additional verification that the model attributes its predictive capability to the physically meaningful variable governing the analytical formulation.

The feature importance results confirm that the normalized radial coordinate constitutes the complete source of predictive information within the generated

dataset. This observation is fully consistent with the mathematical model, in which the relativistic potential depends solely on the dimensionless radial coordinate.

Although feature importance analysis is more informative for datasets containing multiple physical variables, its inclusion here establishes a transparent framework for interpreting machine learning models within a scientific context. Such interpretability analyses become increasingly valuable when extending the present methodology to more complex gravitational systems involving additional physical parameters.

Consequently, the combination of predictive performance evaluation and model interpretability provides a more comprehensive assessment of the applicability of machine learning techniques to computational physics problems.

The contribution of the input variable to the prediction process is illustrated in Figure 7.

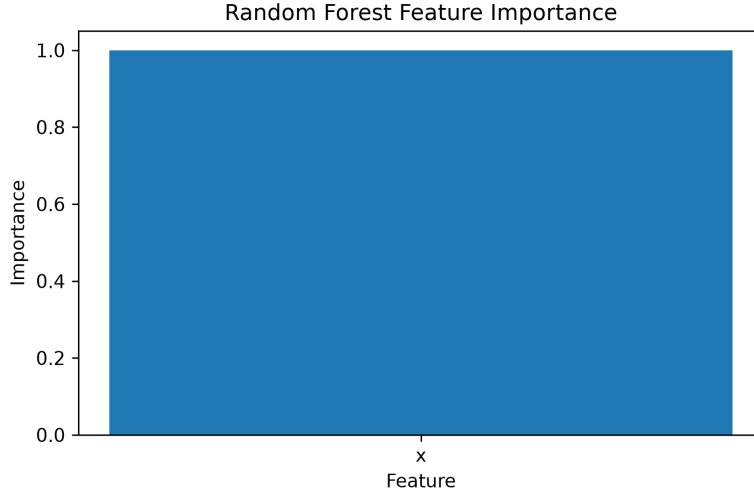


Figure 7: Feature importance obtained from the random forest model. The normalized radial coordinate x is the sole input feature and therefore accounts for the complete predictive contribution.

10 Discussion

10.1 Physics Interpretation

The analytical and computational results provide a quantitative understanding of the relationship between classical and relativistic descriptions of gravitational behaviour.

The error analysis demonstrates that the Newtonian approximation can successfully reproduce the qualitative characteristics of the relativistic model

in the weak-field regime. However, the quantitative difference between the two descriptions becomes increasingly significant as the gravitational field strength increases. This behaviour is consistent with the expected domain of validity of classical gravitational approximations, where relativistic corrections become important only beyond the weak-field limit.

The increasing deviation between the Newtonian and relativistic models in stronger-field regions highlights the limitations of extending classical approximations beyond their applicable range. The transition between relativistic and Newtonian behaviour does not occur at a sharp boundary; instead, it represents a continuous change that can be systematically quantified using the relative error function.

The computational framework developed in this study allows this transition to be investigated by identifying the regions in which the Newtonian approximation satisfies specific accuracy requirements. This provides a measurable interpretation of the validity range of the classical approximation within the considered relativistic potential model.

The results further demonstrate the importance of combining analytical and computational methods in the investigation of physical systems. While the mathematical formulation establishes the theoretical basis of the model, numerical analysis enables detailed examination of its behaviour across different gravitational regimes.

This integration of analytical physics and computational techniques provides the foundation for the machine learning analysis presented in this work, where data-driven methods are evaluated as computational approximations of the established physical relationship.

10.2 Machine Learning Interpretation

The machine learning results demonstrate the capability of data-driven methods to approximate the behaviour of a nonlinear physical system generated from an analytical gravitational model.

The comparison between different algorithms shows that model selection has a significant influence on approximation performance. Simpler methods, such as linear regression, provide an important baseline but are limited in their ability to represent the nonlinear structure of the relativistic potential.

The improved performance of neural networks and random forest regression demonstrates that flexible machine learning architectures can effectively capture complex relationships present within physically generated datasets. These models successfully reconstruct the numerical behaviour of the relativistic potential without requiring repeated evaluation of the original analytical expression during prediction.

However, the purpose of machine learning in this study is not to replace the underlying physical formulation. Since the training data is generated directly from the analytical relativistic potential, the machine learning models are evaluated as approximation techniques rather than independent methods for discovering the governing physical law.

This distinction is important in scientific machine learning, where predictive accuracy must be considered together with physical consistency and interpretability. The present study provides a controlled benchmark in which machine learning models can be compared against an exact theoretical solution, allowing their approximation capability to be evaluated objectively.

The framework demonstrates the potential of machine learning models as surrogate computational tools for nonlinear physical systems. Such approaches may become particularly useful in situations where repeated evaluation of complex models becomes computationally expensive while the underlying physical constraints remain well established.

This concept is consistent with the broader development of scientific machine learning, where data-driven methods are combined with physical knowledge to improve computational efficiency and modelling capability [9].

10.3 Limitations and Future Extensions

Although the results demonstrate the effectiveness of machine learning methods for approximating the relativistic potential, several limitations of the present framework should be acknowledged.

The current dataset is generated from a deterministic analytical model with a single independent variable. This controlled setup allows a clear evaluation of machine learning performance; however, realistic physical systems generally involve multiple parameters, complex boundary conditions, observational uncertainties, and interactions between different physical processes.

Furthermore, the high prediction accuracy obtained by some models is dependent on the quality, distribution, and range of the generated dataset. A model trained within a specific parameter range may not generalize accurately beyond that range without additional training data or appropriate physical constraints.

Future extensions of this work may include incorporating additional physical parameters, investigating more complex gravitational systems, and evaluating machine learning performance using independently generated numerical simulations or observational datasets.

Another important direction is the development of physics-informed machine learning approaches, where governing physical principles are directly incorporated into the learning process. Such methods provide a framework for combining the flexibility of machine learning with the reliability of established physical laws [9].

These extensions would allow the methodology developed in this study to move beyond approximation of a single theoretical relationship toward more complex scientific modelling problems where analytical solutions may be unavailable or computationally expensive.

The present work represents a controlled demonstration of how analytical physics models, numerical analysis, and machine learning techniques can be integrated within a unified computational framework. Although the current investigation focuses on a single relativistic potential, the proposed methodology

provides a foundation for applying similar approaches to broader problems in computational physics.

11 Conclusion

This study presented a combined analytical, numerical, and machine learning framework for investigating a static relativistic gravitational potential and its relationship with the Newtonian approximation.

The theoretical analysis established the mathematical formulation of the relativistic potential and provided a quantitative comparison with the classical model through relative error analysis. The results demonstrated that although the Newtonian approximation reproduces the general inverse-distance behavior of the relativistic model, a systematic quantitative difference remains due to the different asymptotic behavior of the two formulations.

The numerical analysis allowed the deviation between the two models to be investigated across different gravitational regimes. By evaluating the relative error over the selected domain, the study identified the limitations of the Newtonian approximation under the considered formulation and demonstrated how computational methods can provide measurable criteria for approximation validity.

The machine learning component further explored the ability of data-driven models to approximate the relativistic potential from generated physical data. The comparison between linear regression, polynomial regression, neural networks, and random forest regression showed that models with greater capability to represent nonlinear relationships achieved improved prediction performance.

However, the machine learning models were treated as computational approximations rather than replacements for the underlying physical formulation. The analytical model remains the source of the physical relationship, while machine learning provides an alternative method for efficient numerical representation.

Overall, this study demonstrates the potential of integrating theoretical physics, numerical computation, and machine learning for studying nonlinear scientific systems. The developed framework provides a foundation for future extensions involving more complex gravitational models, additional physical parameters, and physics-informed machine learning approaches.

12 Data and Code Availability

The complete source code, datasets, machine learning scripts, generated figures, and supplementary materials associated with this study are publicly available at:

<https://github.com/Dev-Aayush-Kumar/computational-gravity-ml>

This repository contains all computational workflows used to generate the numerical results, machine learning models, datasets, and visualizations presented in this paper.

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I would also like to thank a school friend whose own research publication motivated me to begin this journey. Seeing a fellow student undertake independent research helped me realize that such goals were achievable and encouraged me to pursue my own AI interests with greater confidence and determination.

Finally, I would like to express my profound appreciation to a very special individual whose influence on my personal and academic growth has been immeasurable. At a time when I often underestimated my own potential and struggled with distractions, self-doubt, and uncertainty about my direction, this person consistently encouraged me to aim higher and take my ambitions seriously.

More importantly, this individual demonstrated a level of belief in my abilities that often exceeded my own. Through encouragement, honest conversations, constructive advice, and unwavering support, I gradually began to recognize possibilities within myself that I had previously overlooked. The confidence, discipline, and sense of purpose developed during this period played an important role in enabling me to undertake projects such as the present work.

The research presented in this paper is the result of many months of study, persistence, and effort. However, much of the determination required to begin and continue this journey was strengthened by the support and encouragement received from this individual. Beyond the completion of this project, the influence has extended to broader academic aspirations, personal development, and the pursuit of goals that once seemed beyond reach.

Some contributions to a person's growth cannot be measured through equations, datasets, or performance metrics. Nevertheless, they can leave a lasting impact. For that reason, I offer my sincere gratitude and appreciation for the encouragement, faith, and motivation that helped make this study possible.

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